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[GS PAPER II: GOVERNMENT POLICIES AND IMPLEMENTATION; DEVELOPMENT PROCESSES. • GS PAPER III: AGRICULTURE, ENERGY SECURITY, INFRASTRUCTURE, ENVIRONMENTAL POLLUTION AND BIOTECHNOLOGY.]

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GOBARdhan: Linking Crop Residue Management with India's Energy Security



GOBARdhan turns agricultural residues and organic waste into compressed biogas, supporting cleaner energy, rural livelihoods and India's energy security. Its success depends on reliable feedstock supply, efficient collection networks and remunerative returns for farmers.

GOBARdhan: Linking Crop Residue Management with India's Energy Security

UPSC syllabus relevance: GS Paper II: Government policies and implementation; development processes. **GS Paper III:** Agriculture, energy security, infrastructure, environmental pollution and biotechnology.

India's Waste–Energy Challenge

India faces two interconnected challenges: agricultural residues are often burnt to clear fields, while imported natural gas exposes the economy to international price fluctuations, geopolitical disruptions and foreign-exchange expenditure.

Converting suitable organic waste into **compressed biogas (CBG)** can address part of this challenge by supporting domestic fuel production, waste management and rural livelihoods. However, its success depends on reliable feedstock supplies and commercially viable participation by farmers.

Understanding the Basic Concepts

Concept	Meaning
Crop residue	Plant material remaining after harvesting, including straw, stalks and stubble.
Feedstock	Raw material used in a production process, such as cattle dung or agricultural residue supplied to a biogas plant.
Anaerobic digestion	Breakdown of organic material by microorganisms in the absence of oxygen.
Biogas	Gas produced through anaerobic digestion, containing mainly methane and carbon dioxide.
Compressed biogas	Methane-rich fuel produced by purifying and compressing biogas.

Concept	Meaning
Digestate	Material remaining after anaerobic digestion, which can be processed into organic manure.
Circular economy	An approach that retains the value of resources by converting waste into useful inputs and products.

Production process: Organic waste !’ Anaerobic digestion !’ Biogas !’ Purification and compression !’ CBG

Agricultural residues differ in their suitability for digestion. Fibrous material such as paddy straw may require preprocessing or pretreatment to improve its breakdown.



GOBARdhan: Policy Framework

GOBARdhan—Galvanizing Organic Bio-Agro Resources Dhan— promotes productive utilisation of organic waste. The original initiative was launched in **2018**. Building on this foundation, the Union Cabinet approved an integrated national CBG framework in **August 2026**.

Feature	Details
Outlay	Rs. 23,731 crore
Implementation period	FY 2026–27 to FY 2035–36
Administering ministry	Ministry of Petroleum and Natural Gas
Production objective	Nearly tenfold expansion in domestic CBG production
Support measures	Assured procurement, stable pricing, capital assistance, pipeline infrastructure, credit guarantees and ecosystem development

These measures seek to improve project viability and attract investment across the CBG value chain.

LNG and CBG: Different Supply Chains

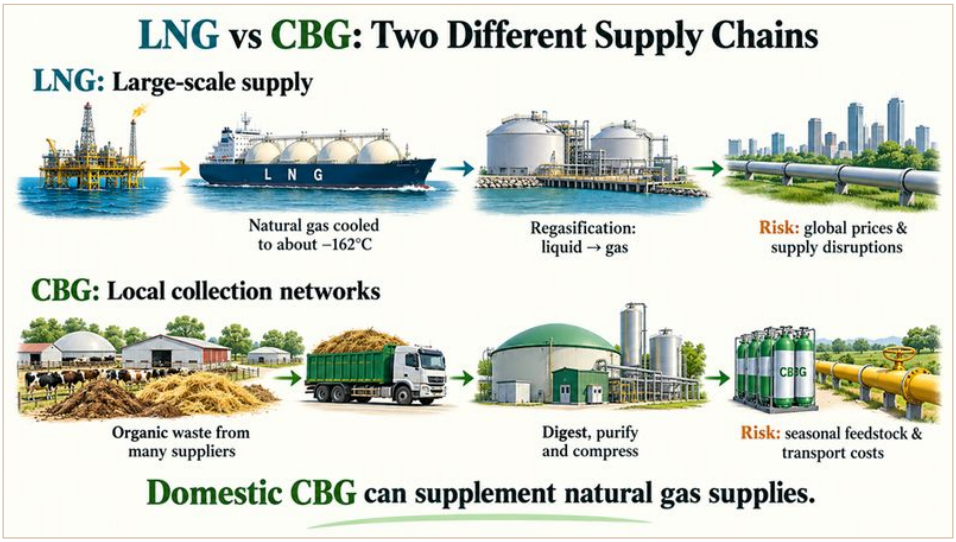
Natural gas is a fossil fuel composed mainly of methane. For transportation by specialised ships, it is cooled to approximately **" 162°C** to produce **Liquefied Natural Gas (LNG)**. At receiving terminals, LNG is converted back into gas through **regasification** and supplied through pipelines.

CBG production, by comparison, depends on collecting organic material from farms, livestock holdings and other local sources.

Aspect	LNG	CBG
Resource base	Fossil natural gas	Renewable organic feedstock
Supply structure	Large shipments and specialised terminals	Numerous, dispersed suppliers

Aspect	LNG	CBG
Infrastructure	LNG carriers, terminals and pipelines	Collection machinery, storage, digesters and gas-distribution facilities
Major risks	Import dependence, global prices and geopolitical disruptions	Seasonal availability, transport costs and irregular feedstock supply

CBG can supplement domestic gas supplies and substitute for natural gas in suitable applications. It does not automatically eliminate the need for imports.



Why Farmers Burn Crop Residue

In some paddy–wheat regions, the interval between paddy harvesting and wheat sowing is approximately **two weeks**. Farmers must clear fields quickly to avoid delaying the next crop.

Burning provides rapid clearance, but contributes to **air pollution, greenhouse gas emissions and nutrient loss**. Alternatives must therefore fit farmers' agricultural schedules and financial circumstances.

Farmers' willingness to supply residue depends on the return remaining after collection and delivery expenses:

Net return = Payment received " Collection, handling, storage and transportation costs

The **factory-gate procurement price** refers to the price paid for feedstock delivered to the plant. A favourable announced price may provide little incentive if delivery costs are high or payments are delayed.

Policy Measures to Reduce Investment Risk

1. CBG Obligation

The **CBG Obligation (CBO)** creates demand for CBG in the **CNG transport and domestic PNG segments** of city gas distribution networks.

The prescribed trajectory is:

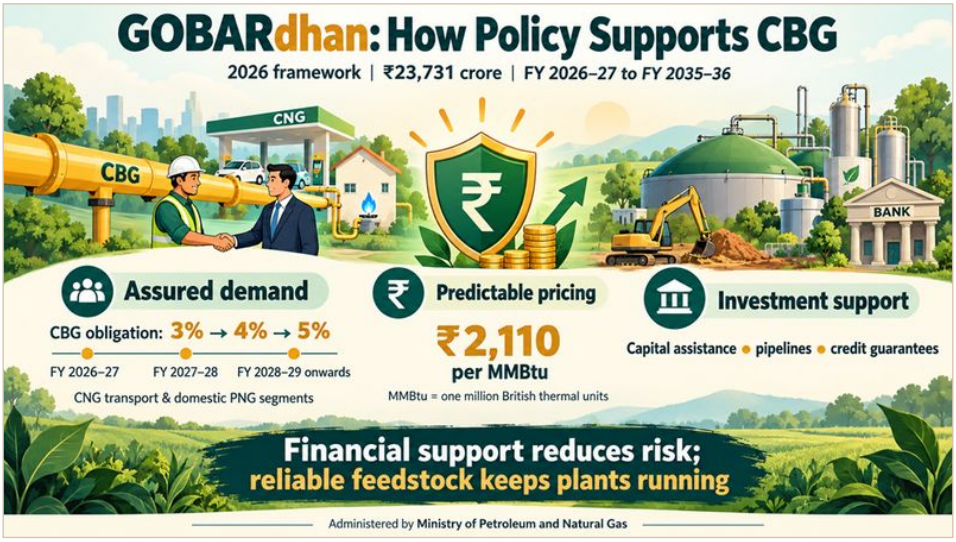
- **3%:** FY 2026–27
- **4%:** FY 2027–28
- **5%:** FY 2028–29 onwards

CNG means compressed natural gas, while **PNG** means piped natural gas.

2. Predictable Pricing and Financial Support

The framework provides an administered CBG price of **Rs. 2,110 per MMBtu**. **MMBtu means one million British thermal units**, a measure of energy content.

Capital assistance, pipeline support and credit guarantees further reduce specified investment and financing risks. Such **de-risking** improves investment conditions; it does not guarantee profitability or continuous feedstock availability.



Major Implementation Challenges

Challenge	Implication
Bulky feedstock	Loose residues occupy substantial space, increasing handling and transport costs.
High moisture content	Transporting wet feedstock also involves moving considerable quantities of water.
Seasonal availability	Residues arrive during harvest periods, whereas plants require regular supplies.
Storage requirements	Year-round operation requires storage capacity and protection against moisture, degradation and fire.
Fragmented suppliers	Coordinating numerous farmers and transport operators increases transaction costs.
Limited mechanisation	Smallholders may lack affordable access to collection equipment.
Underutilised capacity	Feedstock shortages can reduce plant output and raise production costs per unit.

Plants also need **working capital**—funds for routine expenses such as purchasing, transporting and storing feedstock before receiving revenue.



Regional Expansion and Farmer Institutions

Expansion into underserved agricultural regions should follow **local feedstock availability, infrastructure and economic viability**.

Farmer Producer Organisations (FPOs), cooperatives and aggregators can pool material from small farms, organise machinery and transport, negotiate contracts and facilitate payments.

Existing agricultural and dairy networks can support collection. Regions without such institutions require additional capacity building and infrastructure support.

Dedicated Energy Crops: Opportunities and Trade-offs

Energy crops are cultivated specifically for fuel production. **Napier grass**, a fast-growing perennial grass also used as fodder, can serve as biogas feedstock.

Cultivation contracts may provide a more predictable supply alongside seasonal residues. However, their suitability requires assessment of **water demand, cultivation costs, land use and competition with food or fodder production** .

- **Match plant capacity to local resources:** Assess sustainably available feedstock before approving projects.
- **Improve collection:** Expand access to balers, which compress loose residue into manageable bundles.
- **Develop local storage:** Bridge the gap between seasonal harvesting and continuous plant operation.
- **Strengthen farmer institutions:** Support FPOs, cooperatives and transparent aggregation arrangements.
- **Ensure timely payments:** Make residue supply financially attractive and dependable.
- **Diversify feedstock:** Reduce excessive dependence on a single seasonal material.
- **Monitor environmental performance:** Control methane leakage and manage digestate safely.
- **Measure actual outcomes:** Track operational capacity, CBG output, waste utilisation and farmer returns alongside plant registrations.

Conclusion

GOBARdhan connects **agricultural waste management, renewable energy and rural development**. Financial incentives can encourage investment, but sustained success requires efficient local supply chains and meaningful returns for farmers.